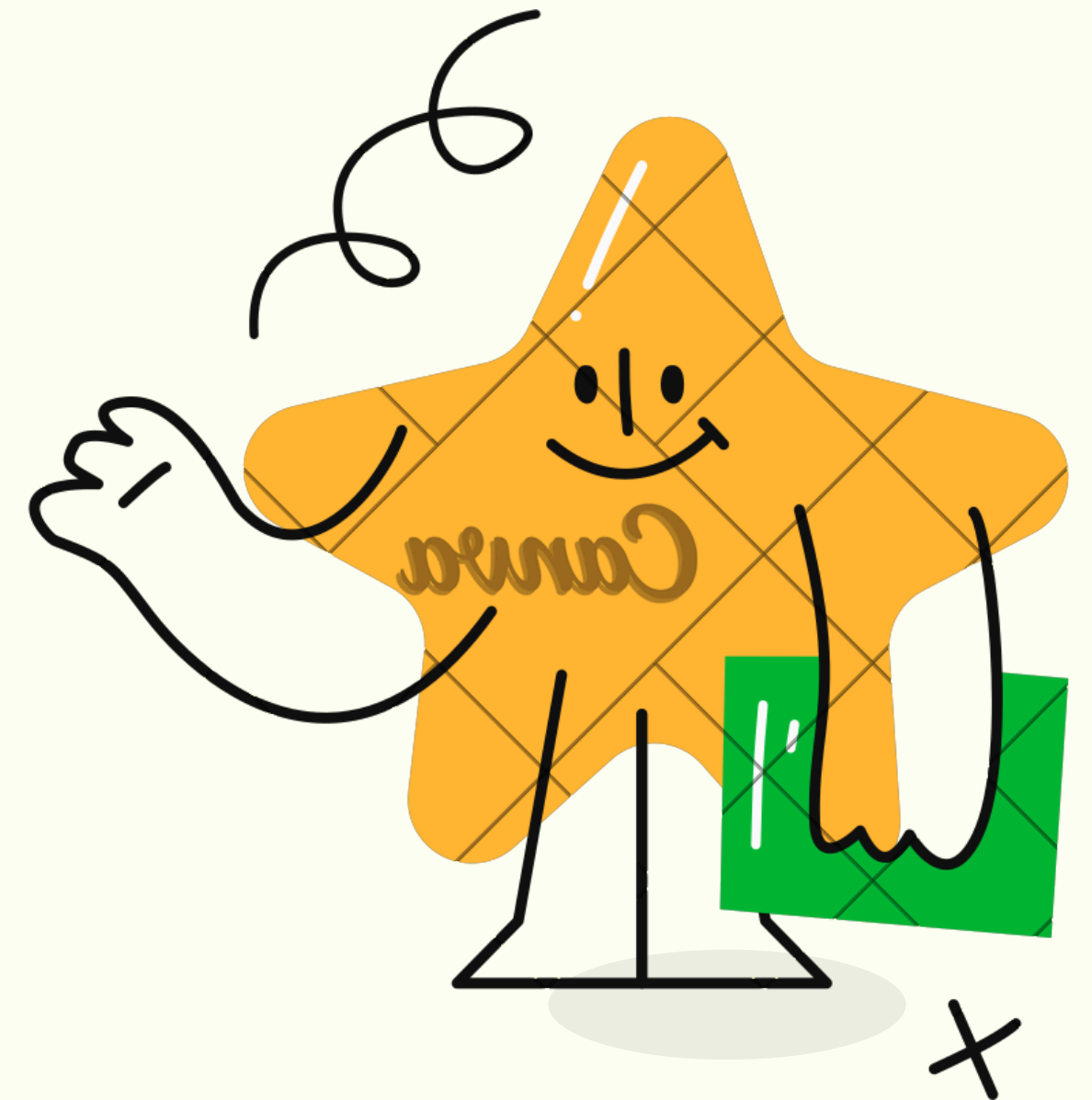
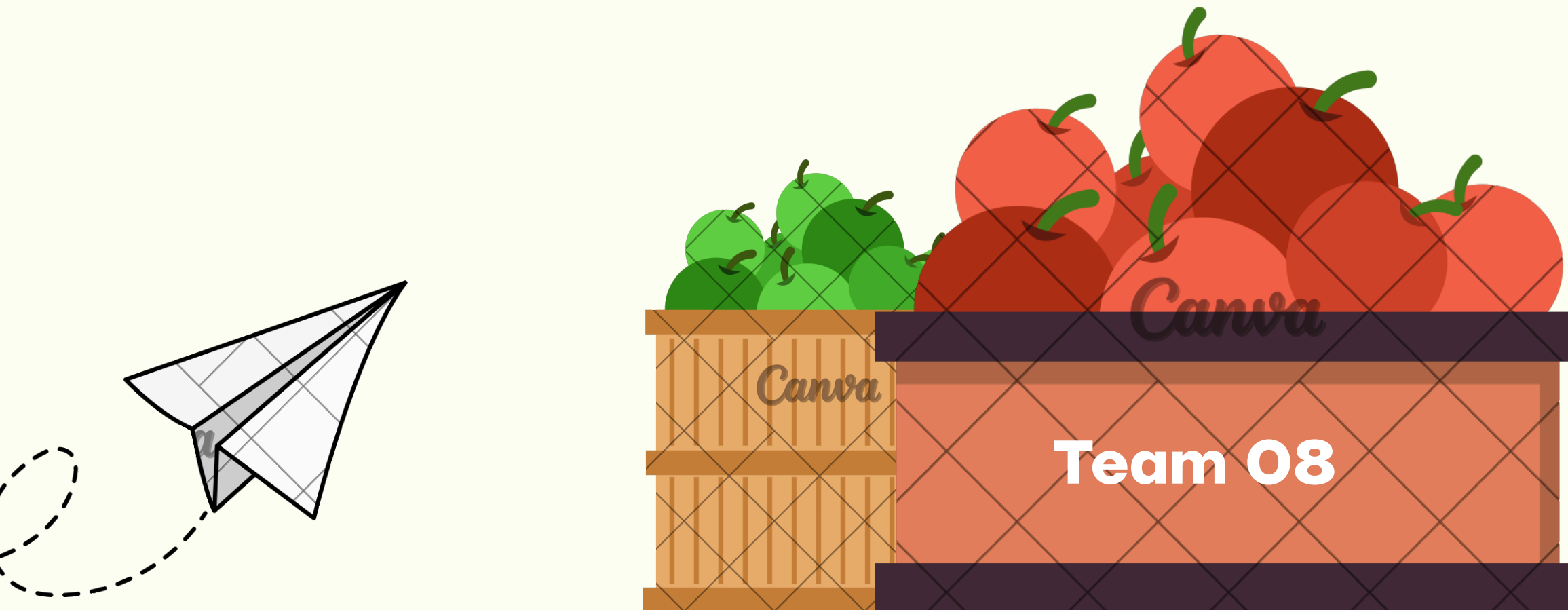


When operations run better, results follow

Sysstat – Performance Monitoring

Collecting, Analyzing, and Reporting System Activity



When operations run better, results follow

Team member

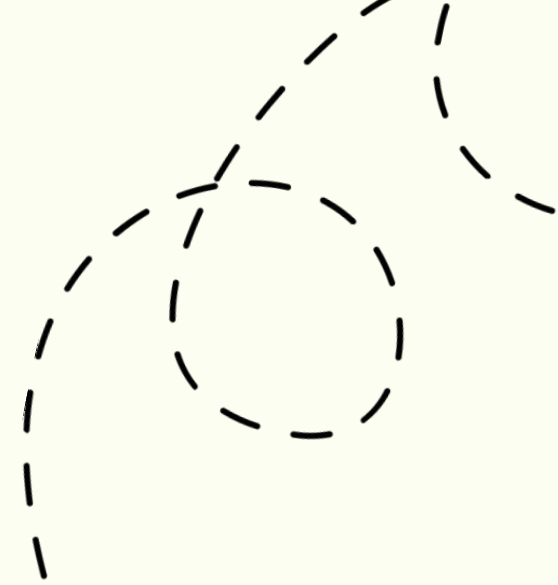
Pham Thanh Dat - 23127170

Huynh Hao Nam - 23127431

Nguyen Van Minh - 23127423

Mai Xuan Hung - 23127372

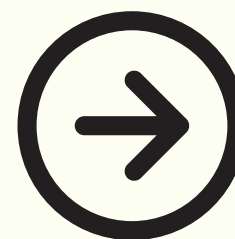
Nguyen Tan Loc - 23127406



Why Do We Need Monitoring Tools?

- A server may appear "slow" — but why?
- Without proper tools, diagnosing root cause is guesswork

Need answers fast



Key questions administrators face



Is the CPU overloaded?



Is the disk I/O bottleneck?



Was the network interface saturated during peak hours?

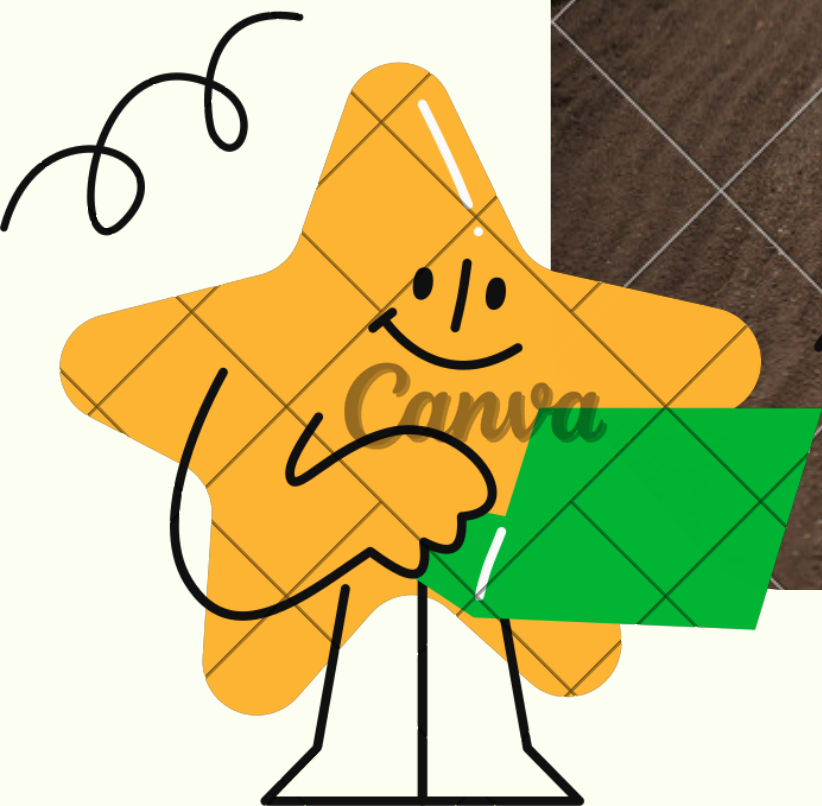
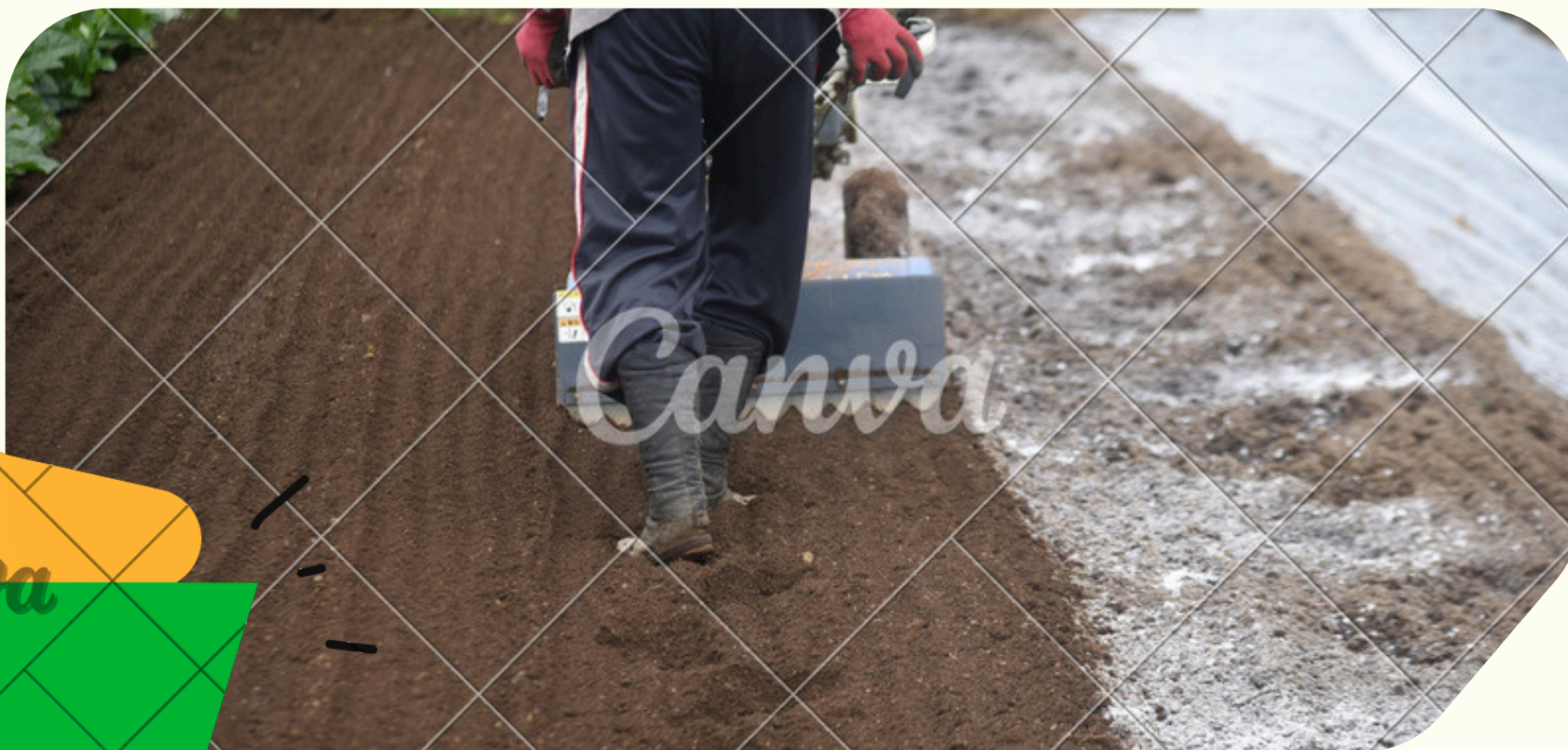


What was wrong with the server at 3 a.m. yesterday that made it run so slowly?



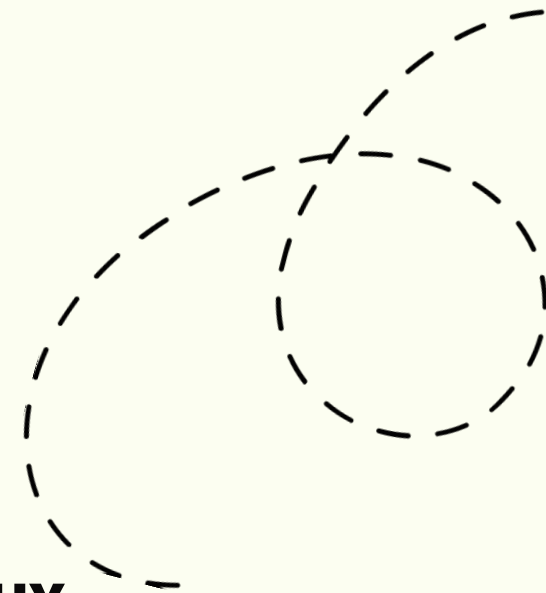
The solution:

Sysstat





What is Sysstat?



- ➔ **Lightweight, standardized Linux monitoring package**
- ➔ **Collects CPU, RAM, Disk I/O, Network, and more**
- ➔ **Used by sysadmins, DevOps, SREs, students**
- ➔ **Goes beyond top/htop — records data over time**

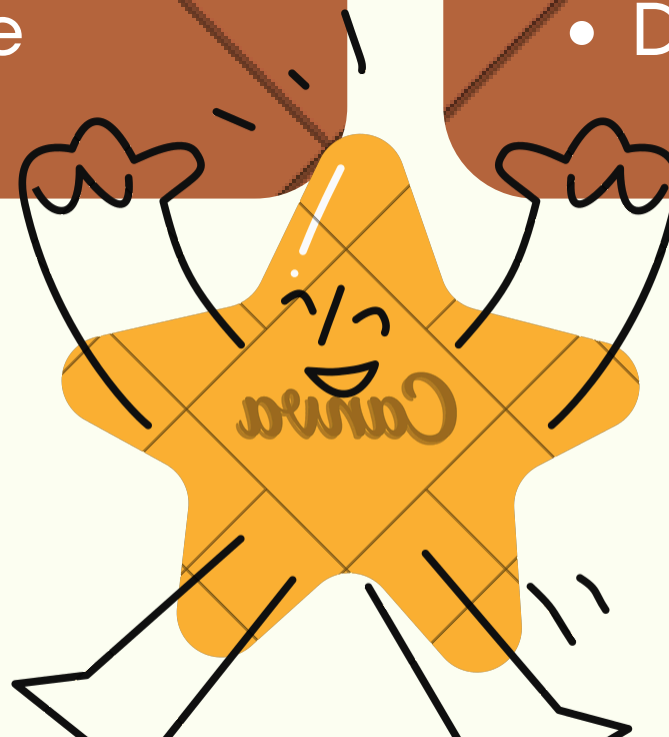
Two Types of Metrics

Real-Time Metrics:

- Show what the system is doing **right now**
- Current CPU utilization %
- Current disk read/write throughput
- Current memory & network usage

Historical Metrics:

- Statistics **recorded & stored** over time
- CPU usage at 2:00 PM yesterday
- Memory trend over the last 24 hours
- Disk activity during last night's backup



Key Commands



sar - System Activity Reporter



iostat - Input/Output Statistics



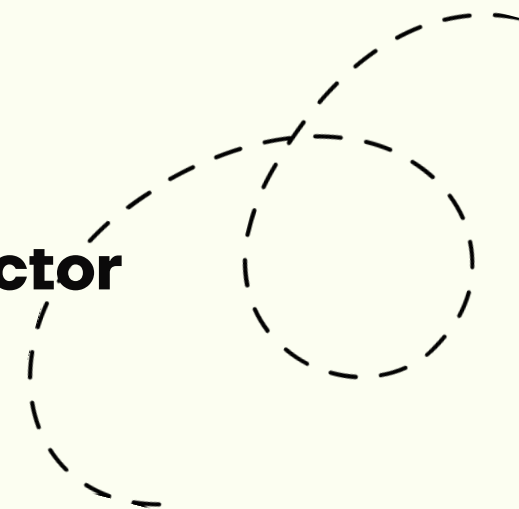
mpstat - Multiple Processor Statistics



pidstat - Processor ID Statistics



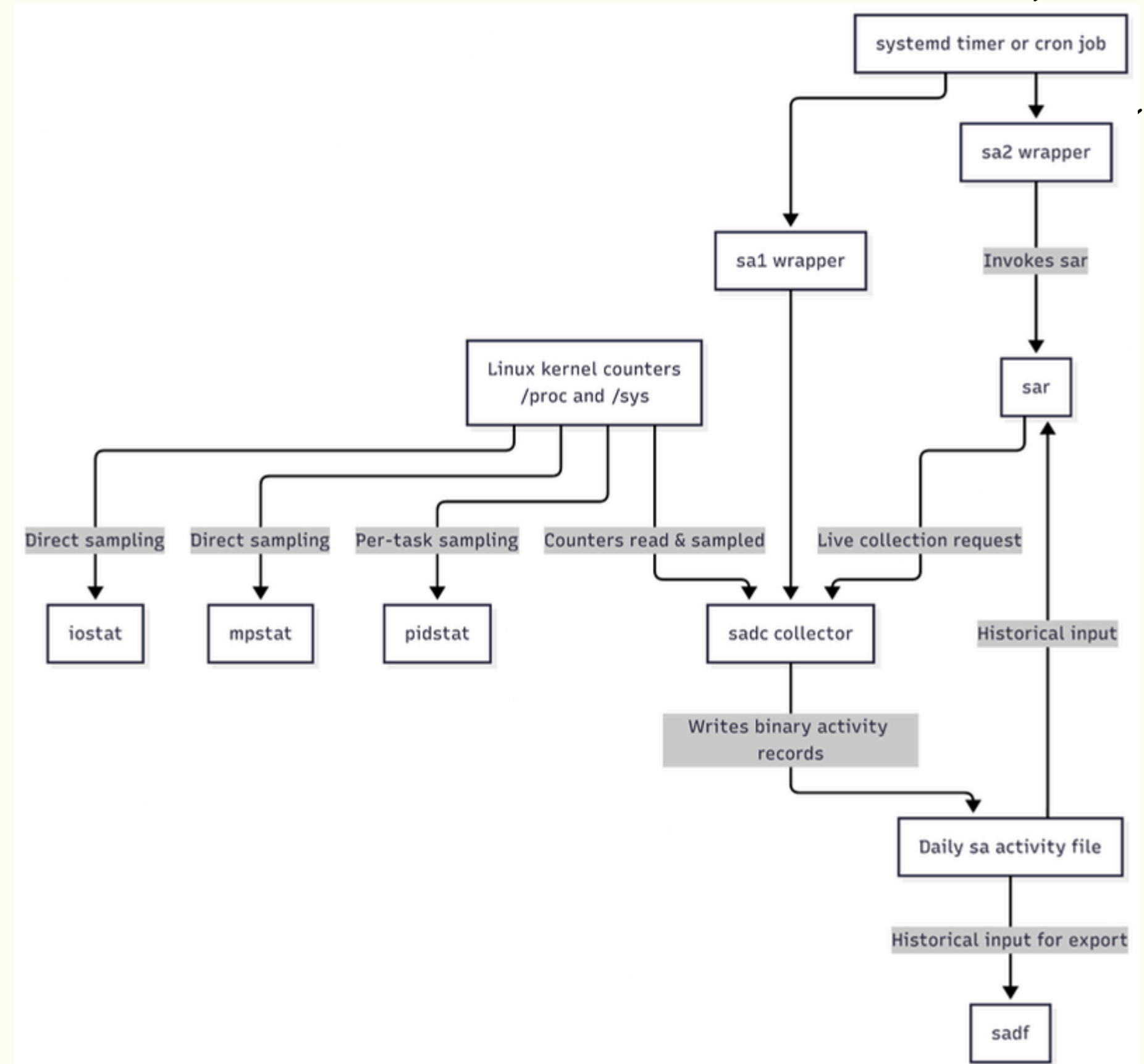
sadc - System Activity Data Collector



How Sysstat Works

Live reporting: `iostat`, `mpstat`, `pidstat` read kernel counters directly

Historical reporting: `systemd` timer → `sa1` → `sadc` → daily file → `sar`



Step 1 - Install

```
sudo apt update  
sudo apt install sysstat
```

Step 2 - Enable Collection

```
sudo dpkg-reconfigure sysstat
```

OR manually:

```
sudo nano /etc/default/sysstat
```

```
sudo systemctl restart sysstat
```

Installing & Enabling Sysstat

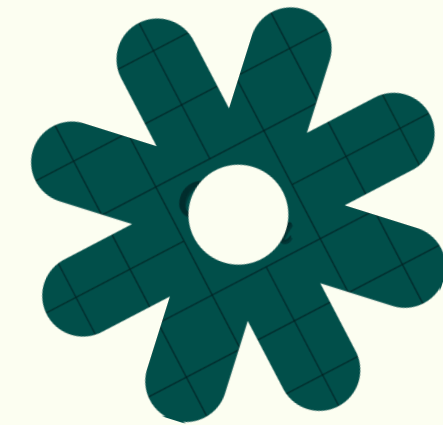
Verify Install

```
sar -V  
command -v sar iostat mpstat \\  
pidstat
```


Configuring Timers & Data Collection

ENABLE SYSTEMD TIMERS

```
sudo systemctl enable --now sysstat-collect.timer  
sudo systemctl enable --now sysstat-summary.timer
```



CUSTOM INTERVAL

```
sudo systemctl edit sysstat-collect.timer  
sudo systemctl daemon-reload  
sudo systemctl restart sysstat-collect.timer
```



Configuring Timers & Data Collection



KEY CONFIG —
`/etc/sysstat/sysstat`



Setting	Purpose
SADC_OPTIONS	Activities to record
HISTORY	Days of data to retain
COMPRESSAFTER	Age at which old files are compressed

Verifying Data Collection Works

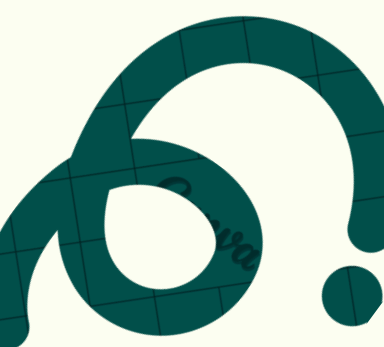
CHECK DATA FILES

```
ls -lh /var/log/sysstat/ # e.g. sa17
sar --sadc               # which collector?
sar -u                  # CPU history
sar -d                  # disk history
```

WHAT TO EXPECT

- File like `sa25` appears in `/var/log/sysstat/`
- `sar -u` shows CPU usage history
- `sar -d` works only if `SADC_OPTIONS` includes `-S DISK`

Benefits & Limitations



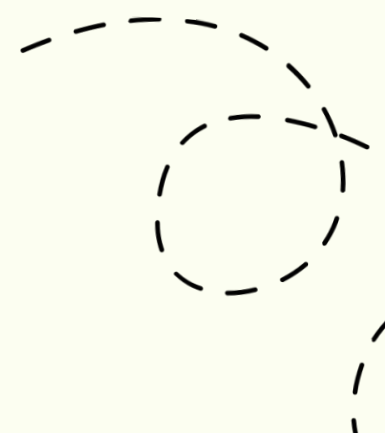
Benefits

- Very lightweight
- No dependencies
- Live + Historical data
- Exportable data (CSV / JSON)



Limitations

- 10-min sampling interval
- No per-process history
- No alerts or graphs
- Single machine only





When to use Sysstat?

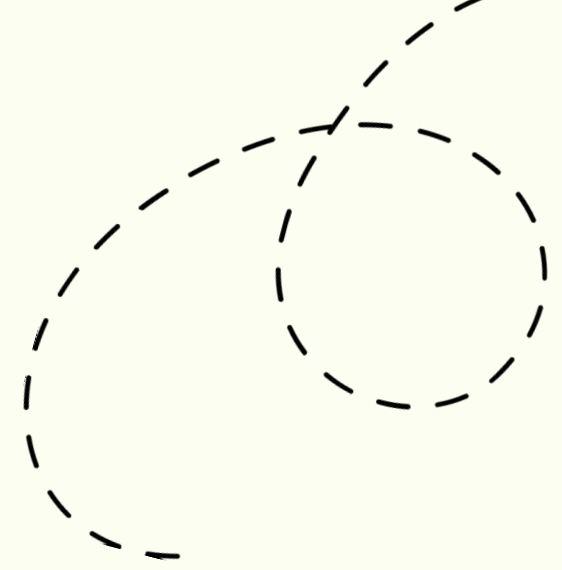
✓ Good fit

- Need to **look into the past**
- Want to **find the bottleneck**
- Machine is **small or offline**
- Want to **compare normal vs. bad days**

✗ Not the Right Tool

- Need **live graphs** or dashboards
- Need **threshold alerts**
- Need to monitor **many servers at once**
- Need to catch a **short-lived process by name** → use **atop**

Alternatives



TRADITIONAL CLI

top / htop / btop

vmstat

pcp-dstat

ADVANCED LOCAL

atop

ENTERPRISE / GUI

glances

netdata

Prometheus + Grafana



Mistakes and Misunderstanding



First Line Anomaly

- First output = average since boot
- Analyze the following samples only.



NVMe %util Misconception

- 100% %util $\neq$ Disk bottleneck
- Check await and throughput instead.

sar shows no data

Enable **sysstat** service before using sar.



Mistakes and Misunderstanding

Reading History Logs

- ❌ `cat /var/log/sysstat/saXX`
- ✅ `sar -f /var/log/sysstat/saXX`

Interval vs Count

Correct syntax:

- `command [interval] [count]`

High %iowait

- Usually indicates Disk I/O bottleneck
- Not necessarily a CPU issue.

High Load Average

- Includes processes waiting for Disk/
Network I/O
- High load $\neq$ High CPU usage.



Team 8

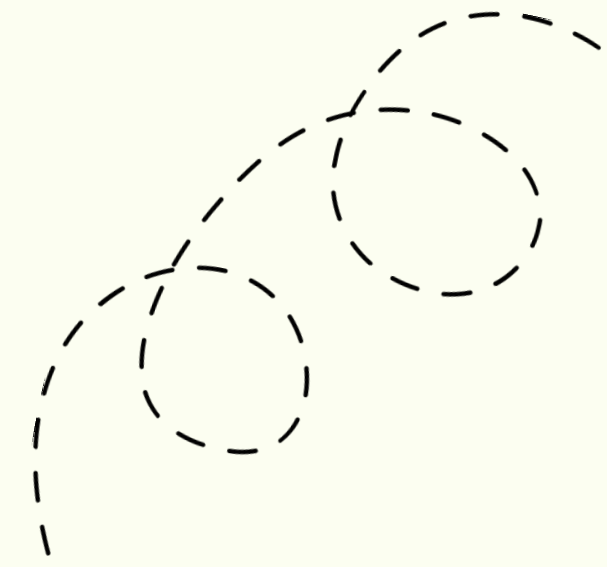
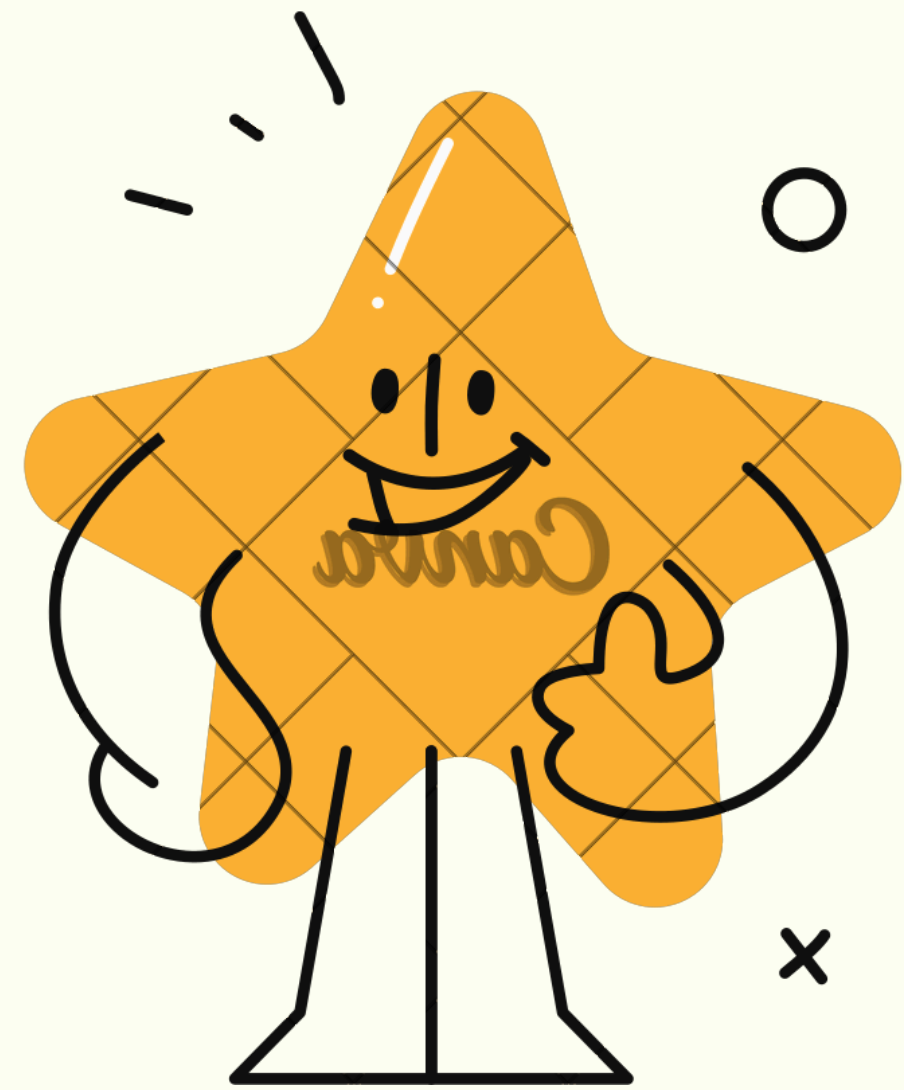
Demostration



Team 8

Thank you so much

Reference: Sysstat github repo





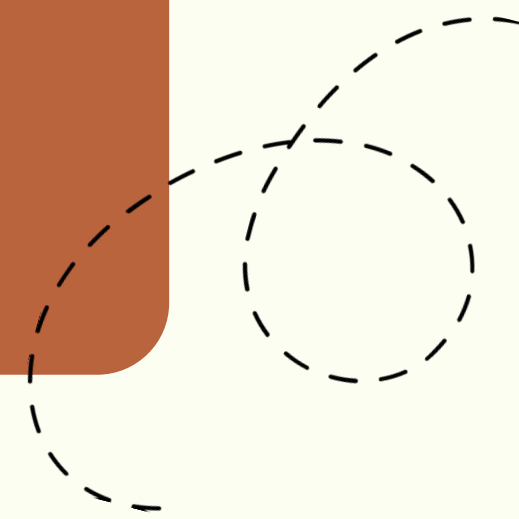
Who It's Designed For

Medium To Large Farms



**Commercial farming operations**

**Plantation and estate farms**



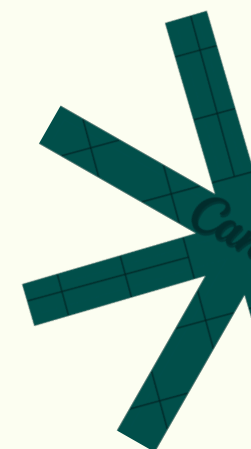
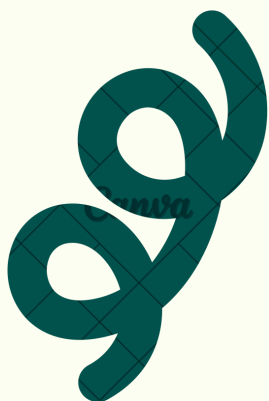
Startup Presentation

Inefficient Farm Operations

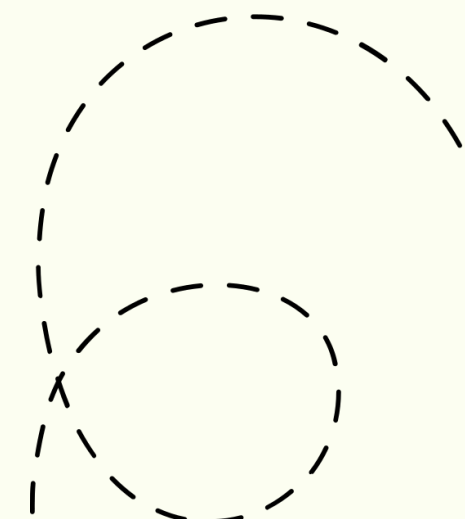
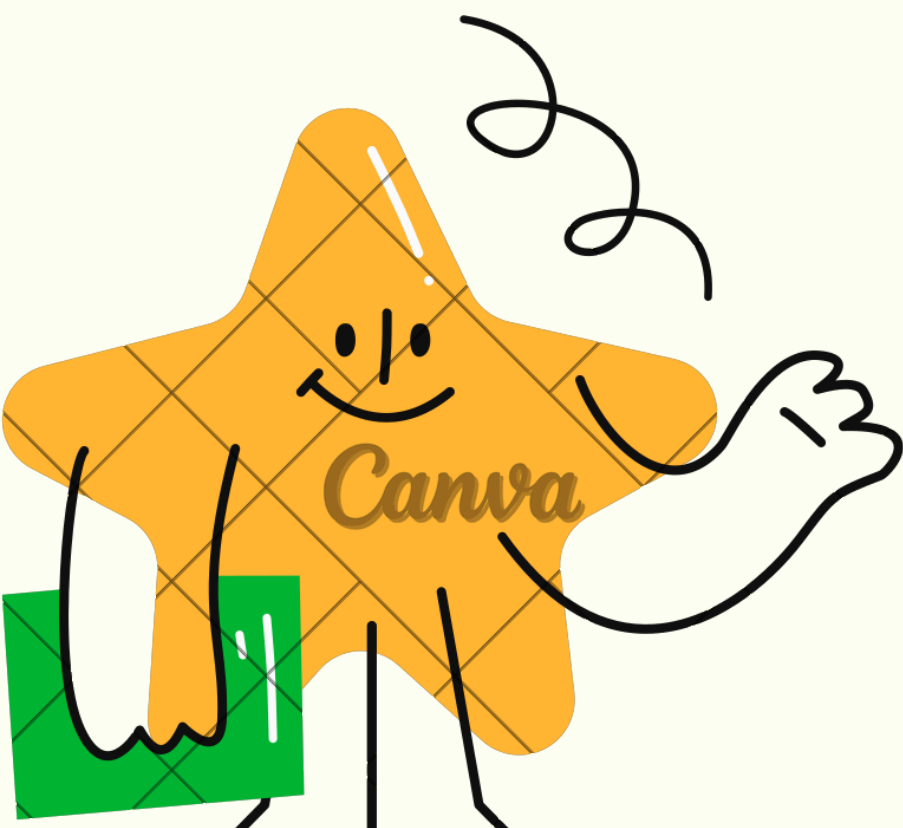
Many farms struggle with fragmented operations, leading to wasted resources, higher costs, and reduced productivity.

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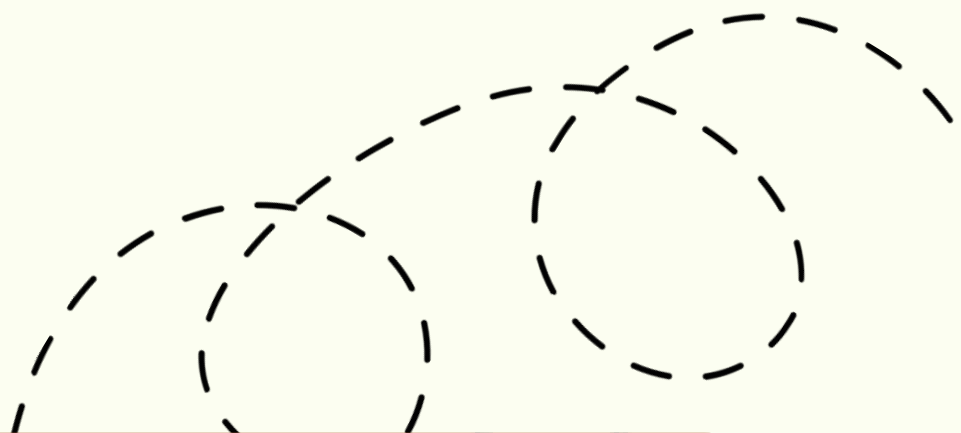


Demonstration





Startup Presentation





Workforce tracking and scheduling



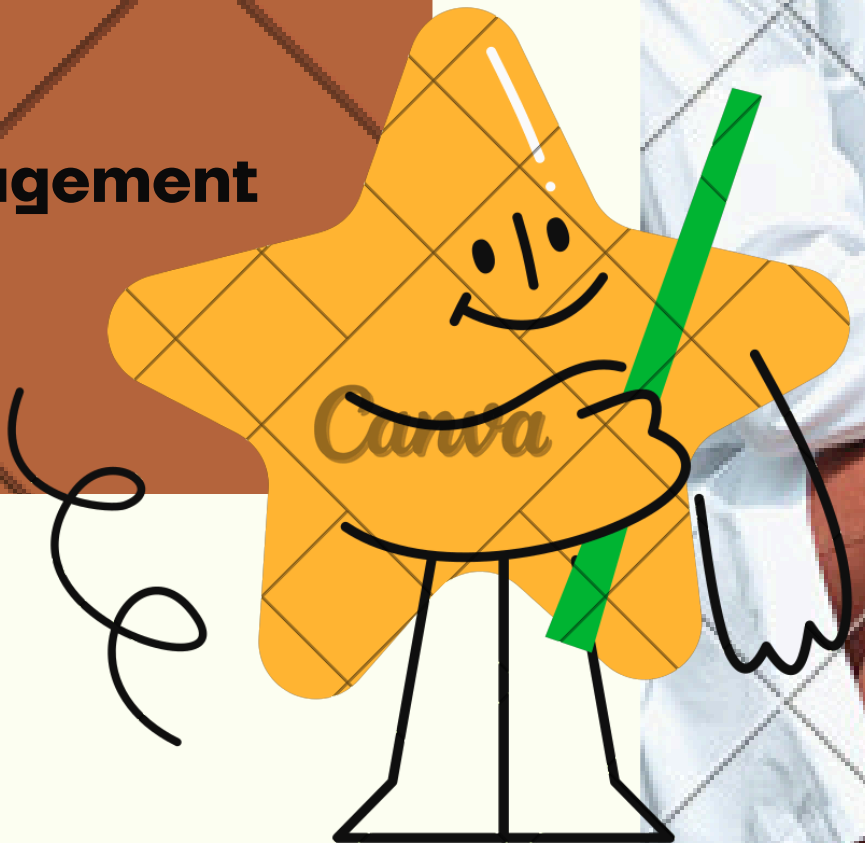
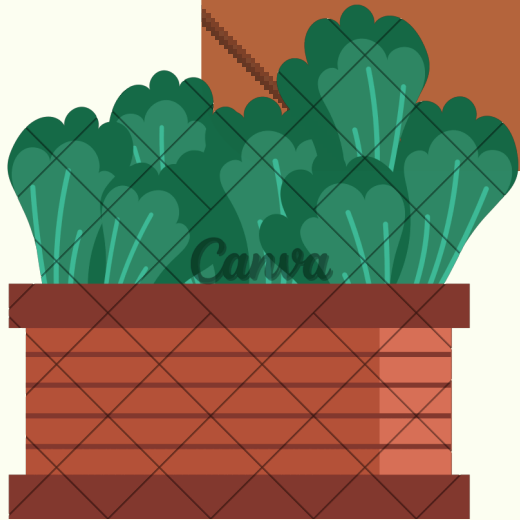
Harvest logging and reporting



Centralized operational dashboard



Inventory management and control



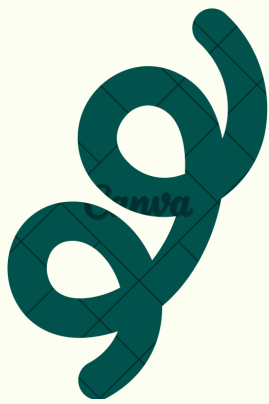
Startup Presentation

What is Sysstat?

➔ No Clear Visibility

Without centralized management, tracking labor, inventory, and harvest data becomes inconsistent and error-prone





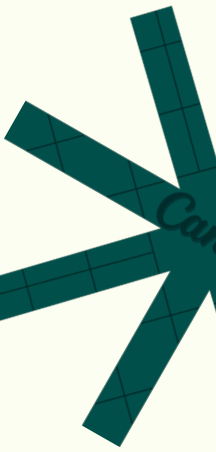
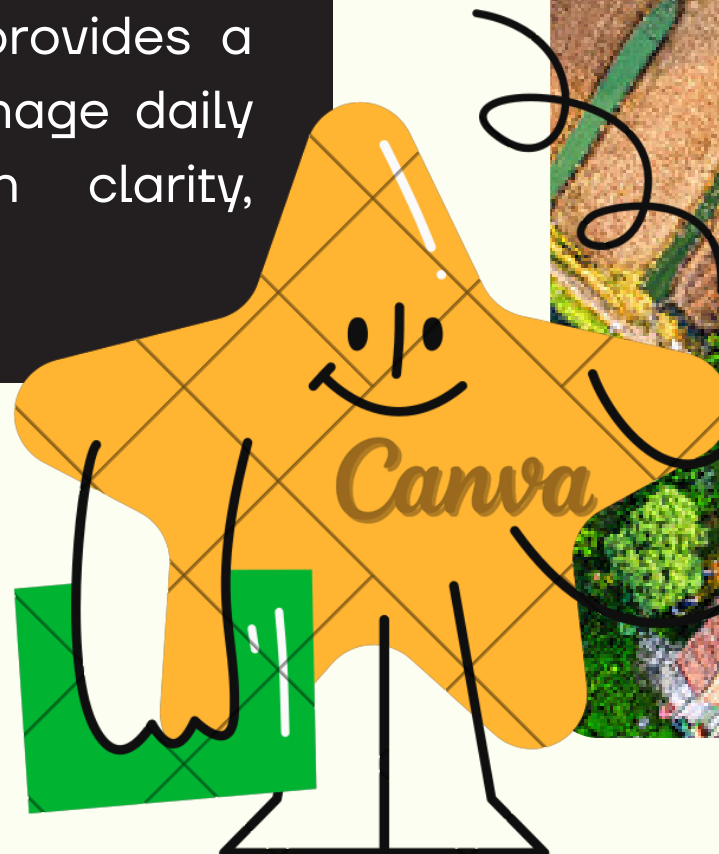
Startup Presentation

Integrated Farm Management Software



Fradel and Spies Co provides a unified platform to manage daily farm operations with clarity, accuracy, and efficiency

www.reallygreatsite.com





**Better inventory
utilization**



**Higher overall
productivity**



**Predictable performance
metrics**



Why It Delivers Impact

Lower Costs Higher Output



Startup Presentation

Optimizing Farm Performance

With better operational control, farms gain efficiency, transparency, and the ability to scale sustainably

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